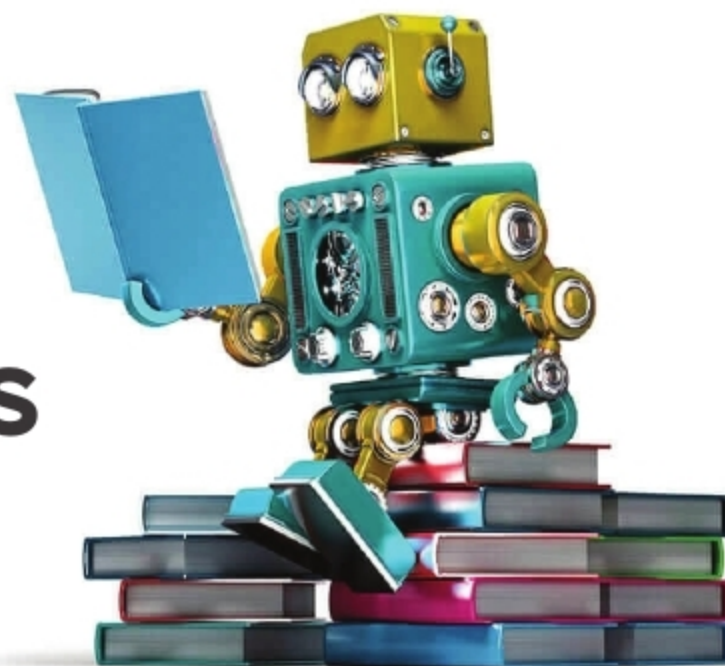


Machine Learning Basics for NÓÖBS



In simple layman terms, machine learning helps in making machines acquire human-like behaviour, while retaining their original capabilities. Here's a quick introduction to the subject.

Machine learning (ML) is a fascinating field in computer science wherein the emphasis is to meet the challenge of the Turing test. The Turing test is focused on providing sufficient intelligence to machines so as to make it difficult for anyone to distinguish between humans and machines. CAPTCHAs were introduced to avoid illegal intrusions into critical online systems by machines impersonating humans. With the advent of improved and more accurate learning algorithms, it seems that in the near future, machines will start outperforming human beings in intelligence.

Arthur Samuel is known as the father of ML. In the 1960s, he visualised a computer that could learn the moves of checker-board games through training, and once sufficient experience was gained it would be able to think by itself and plan the next moves accordingly, to ultimately win games against humans. He turned this thought into reality and we are today witnessing how rapidly this field is emerging. He defined ML as, "The field of study that gives computers the ability to learn without being explicitly programmed."

The need for machine learning

Humans are naturally good at recognition. Early in life, a baby starts recognising her mother's voice and face,



Figure 1: How a machine passes the Turing test (Image source : <http://www.artificialbrain.xyz/can-a-computer-talk-like-a-human/>)

while children soon start identifying various categories of animals, birds or vehicles through observations and a little experience. On the other hand, machines are very powerful in processing and in rapid computation when the input is data intensive. They can keep on doing the same task repeatedly, with the same efficiency, without getting exhausted, which is not possible in the case of humans. Human performance deteriorates over time, when working without a break. The only limitation for machines is that they have very poor recognition capabilities, so if somehow they can be trained like humans, they can start performing better over time and with experience.

Many complex tasks that are really challenging for humans can be automated. If the merits of both humans and machines can be combined, this can result in great performance. It is impossible as of now to equip humans with machine-like capabilities, so an easier option is to make machines learn recognition and gain the other capabilities that humans have. This is how the field of machine learning has evolved (Figure 2).

In the scenario we just discussed, machines can serve humans well and reduce the effort required by us in applications such as biometric authentication, sentiment analysis, recommender systems, anomaly detection, navigation and many others.

Machine learning paradigms

There are three paradigms of ML algorithms, which are discussed in the following sections.

Supervised learning: When data comes with labelled examples stating the class or category of each example, this is called the supervised learning approach. Machines are trained under supervision by providing sufficient examples with class label information. This is learning from examples.

- **Classification** is a supervised learning strategy in which a model is built by providing some training examples; then